



LESSON - 17 & 18

SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- UNION, UNION ALL, MINUS, INTERSECT
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

SubQuery

In SQL, a **Subquery** (also known as an inner query or nested query) is a query within another query. It is enclosed in parentheses and can be used in SELECT, INSERT, UPDATE, or DELETE statements. Subqueries help perform complex queries by breaking them into smaller, more manageable parts.

Types of Subqueries:

- Single-Row Subquery
- Multi-Row Subquery
- Multi-Column Subquery
- Correlated Subquery
- Nested Subquery
- Scalar Subquery
- Inline View

Single-Row Subquery.

Definition: Returns only one row and one column as the result.

Usage: Typically used with operators like =, <, >, <=, >=, and <>.

Syntax / Example :

```
SELECT name, salary
```

```
FROM employees
```

```
WHERE salary > (SELECT AVG(salary) FROM employees);
```

Multi-Row Subquery.

Returns multiple rows from one column.

Used with operators like IN, ANY, ALL

Syntax / Example :

SELECT name

FROM employees

WHERE department_id IN (SELECT department_id FROM departments

WHERE location = 'New York');

Multi-Column Subquery.

Returns more than one column.

Usually used in IN or as a set of columns for comparison.

Syntax / Example :

```
SELECT name, salary
```

```
FROM employees
```

```
WHERE (department_id, job_id) IN (SELECT department_id, job_id FROM  
employees WHERE salary > 10000);
```

Correlated Subquery.

Refers to columns in the outer query, and is evaluated once for each row processed by the outer query.

Syntax / Example :

```
SELECT e1.name, e1.salary  
FROM employees e1  
WHERE e1.salary > (SELECT AVG(e2.salary) FROM employees e2  
WHERE e1.department_id = e2.department_id);
```

Nested Subquery.

A subquery within another subquery.

Syntax / Example :

SELECT name

FROM employees

WHERE department_id = (SELECT department_id

FROM departments

WHERE location = (SELECT location

FROM locations

WHERE city = 'New York'));

Scalar Subquery.

Returns exactly one value (one row and one column). It can be used wherever an expression is valid.

Syntax / Example :

```
SELECT name, (SELECT MAX(salary) FROM employees) AS max_salary  
FROM employees;
```

Inline View

A subquery in the FROM clause that acts like a temporary table or view.

Syntax / Example :

```
SELECT dept_name, avg_salary
FROM (SELECT department_id, AVG(salary) AS avg_salary
      FROM employees
      GROUP BY department_id) AS dept_avg
JOIN departments ON dept_avg.department_id =
departments.department_id;
```

Comparison of Subquery Types:

TYPE	RETURNS	USED WITH	EXAMPLE OPERATORS	TYPICAL USE CASE
Single-Row	One row, one column	SELECT, WHERE	=, <, >, <=, >=	Comparing a value with a single result
Multi-Row	Multiple rows, one col	WHERE, HAVING	IN, ANY, ALL	Checking for existence within multiple values
Multi-Column	Multiple rows, columns	WHERE, IN	IN, EXISTS	Matching multiple columns with a set of values
Correlated	Varies per row	SELECT, WHERE	N/A	Row-by-row comparison with outer query values
Nested	Subquery within subquery	Any clause	N/A	Complex filtering or condition checks
Scalar	One value (one row, one col)	SELECT, WHERE, ORDER BY	=, <>, etc.	Using a single value within an expression
Inline View	Temporary result set	FROM	N/A	Acting as a derived table for further querying